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DS4300
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Homework 3: Document Stores

Nutrition is one of the most important aspects to human health and longevity. The FDA Food recall enforcement is essential for protecting public health by removing products that are either defective or potentially harmful from the market and correcting the problem from there. In this project, not only did we gain a better understanding of the data model, performance, and query syntaxes of MongoDB but also set up an API that can help users analyze and observe overall trends of recalls from the past 20 years.

Our JSON-formatted dataset was found from open.fda.gov. The data consists weekly-updated food recall records from the FDA Recall Enterprise system (RES) from 2004 to present. The JSON consists of 26 fields, including information on the company/firm (id, address), the product (id, quantity, type), and the recall (reason, classification). Once the dataset was downloaded, we were able to extract the JSON file using Python and import it into MongoDB using Bash.

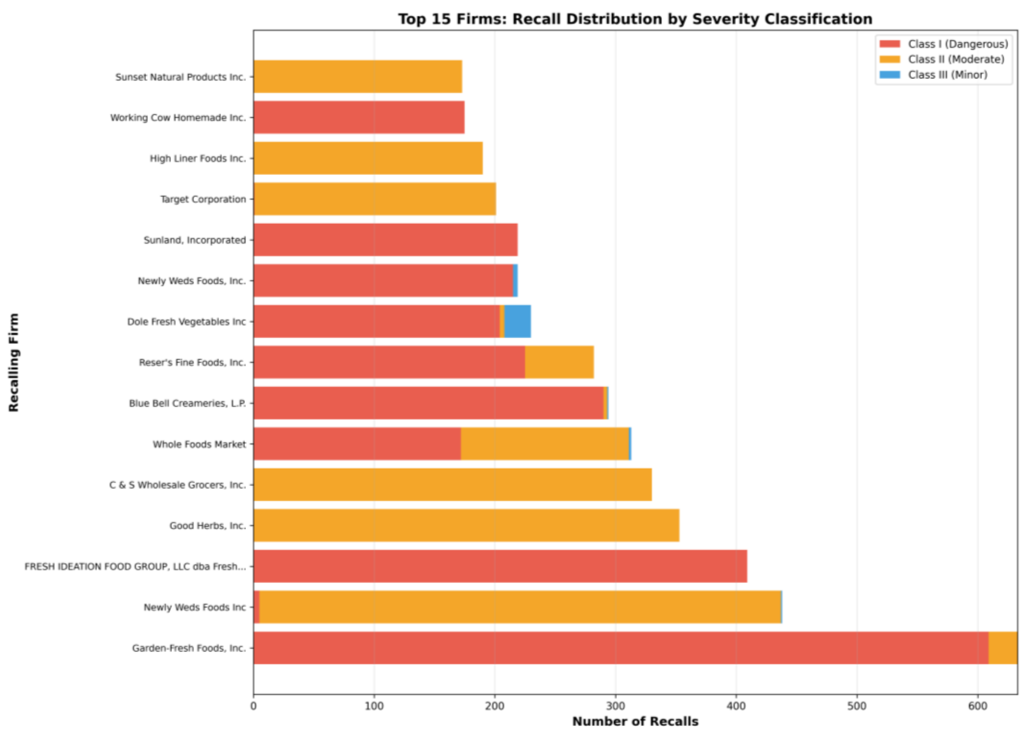

Figure 1: Top 15 Firms - Recall Distribution by Severity Classification

Garden-Fresh Foods, Inc. leads with 633 total recalls, predominantly classified as Class I (dangerous). The stacked bar visualization reveals significant variation in recall severity across firms. While some companies, such as Good Herbs and C&S Wholesale Grocers show primarily Class II (moderate) recalls, others like Blue Bell Creameries and

Fresh Ideation Food Group demonstrate higher proportions of the more dangerous Class I violations. This firm-level breakdown reveals that having higher recall counts doesn't necessarily represent dangerous product recalls but instead reflects larger production volumes and potentially more compliance.

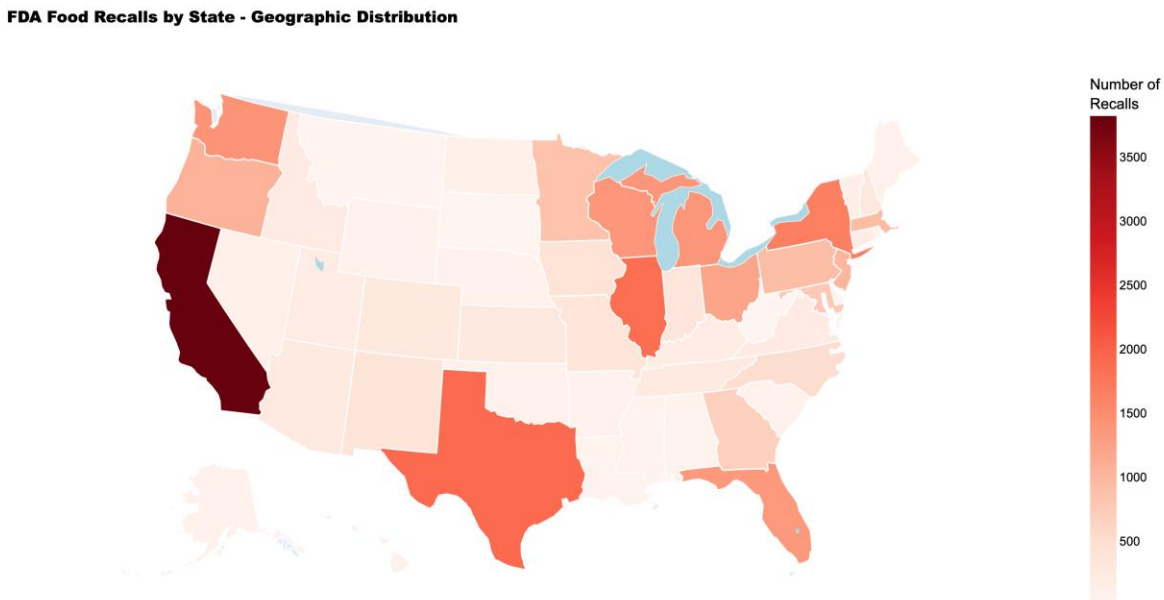

Figure 2: Geographic Distribution of FDA Food Recalls

The choropleth map reveals the geographic concentration of FDA food recall enforcement. California has the highest count with 3,822 recalls (13.5% of all recalls), shown in the dark red. Texas (1,907 recalls) and Illinois (1,868 recalls) follow as the second and third most-affected states. High recall numbers could indicate stronger enforcement rather than worse food safety. California's high count could be reflected by a large food industry, strict state regulations, and active consumer protection agencies. When normalized by food production volume, California's recall rate might align with those from other states, suggesting the geographic pattern reflects industry distribution more than safety failures.

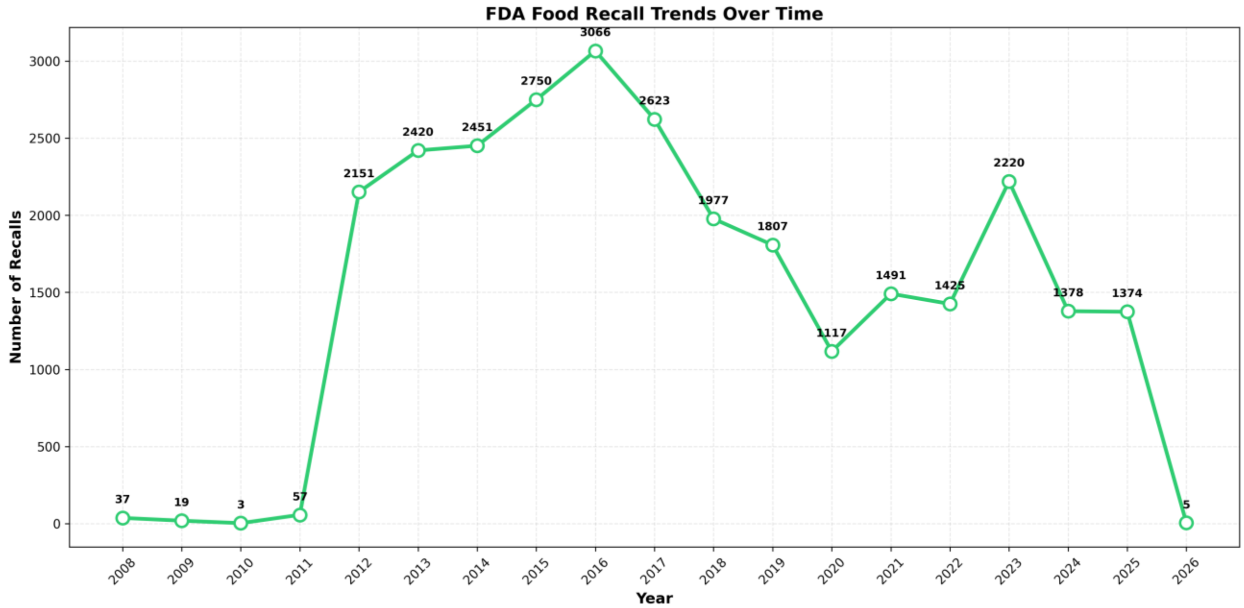

Figure 3: Temporal Trends in FDA Food Recalls (2008-2026)

The temporal trend reveals three distinct eras in FDA food recall enforcement. From 2012 to 2016, the annual recall count increased dramatically from 2151 to 3066. This could be due to the 2011 Food Safety Modernization Act (FSMA), which gave the FDA expanded authority for preventive controls and mandatory recalls. From 2017 to 2020, the annual recall count decreased from 2623 to 1117. This could be because of the COVID-19 pandemic, where FDA inspections, productions, and regulation priorities reduced significantly. From 2022 to 2025, the annual recall count rose back to around the range of 1400-2200. This could be due to post-COVID-19 pandemic, when regulations and productions were back to normal standards. This graph shows how a significant event can disrupt regulatory oversight, raising questions about undetected food safety issues during reduced inspection periods.

**Insights from the API function calls are provided on the run_mongo_api.py file